Lamb Shift Dependence on Resonator Mode Structure: $\lambda/4$ versus $\lambda/2$ in Circuit QED

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We propose a direct comparison of the Lamb shift experienced by a transmon qubit coupled to a $\lambda/4$ (quarter-wave) versus a $\lambda/2$ (half-wave) superconducting coplanar or coaxial resonator. The $\lambda/2$ resonator supports all harmonics $(f, 2f, 3f, \ldots)$, while the $\lambda/4$ supports only odd harmonics $(f, 3f, 5f, \ldots)$. The vacuum fluctuations of the "missing" even modes contribute to the qubit Lamb shift in the $\lambda/2$ case but not in the $\lambda/4$ case. We compute the expected difference $\Delta(\delta f) \sim 1\,\mathrm{MHz}$ for typical circuit QED parameters and show it is well above the measurement floor ($\sim 1\,\mathrm{Hz}$ with standard techniques). This measurement provides a direct quantification of multi-mode vacuum fluctuation contributions to qubit frequency, relevant to understanding frequency stability and residual shifts in multi-mode quantum processors. No new hardware is required: both resonator types and transmon qubits are standard components in superconducting circuit laboratories.

MOTIVATION

Superconducting transmon qubits are routinely coupled to microwave resonators in the dispersive regime [1, 2]. The qubit frequency acquires a shift from virtual photon exchange with each resonator mode (the Lamb shift):

$$\delta f = \sum_n \frac{g_n^2}{f_q - f_n}, \tag{1}$$

where g_n is the coupling to mode n, f_q the bare qubit frequency, and f_n the n-th resonator mode frequency [3, 4].

In multi-mode environments—a regime of growing importance for quantum processors with bus resonators, multiplexed readout, and 3D cavities [5, 6]—the sum in Eq. (1) extends over many modes, and higher modes make non-negligible contributions. Accurately predicting and controlling these multi-mode shifts is essential for qubit frequency targeting and cross-talk minimization.

Despite this practical importance, **no experiment has directly compared the Lamb shift in resonators with different mode structures**. We propose the simplest such comparison: $\lambda/4$ versus $\lambda/2$.

MODE STRUCTURE

A $\lambda/2$ resonator (both ends shorted, or both open) supports all integer harmonics:

$$f_n^{(\lambda/2)} = n\, f_0, \quad n = 1, 2, 3, 4, \ldots \tag{2}$$

A $\lambda/4$ resonator (one end shorted, one end open) supports only odd harmonics:

$$f_n^{(\lambda/4)} = (2n{-}1)\, f_0, \quad n = 1, 2, 3, \ldots \tag{3}$$

Both have the same fundamental frequency f_0 if the $\lambda/4$ resonator is half the physical length of the $\lambda/2$ resonator (accounting for the factor of 2 in the standing-wave condition).

The two resonators share all odd-harmonic modes; they differ only in the presence (absence) of even harmonics in the $\lambda/2$ ($\lambda/4$) case.

PREDICTED LAMB SHIFT DIFFERENCE

The Lamb shift difference between the two configurations isolates the even-mode vacuum contribution:

$$\Delta(\delta f) = \delta f^{(\lambda/2)} - \delta f^{(\lambda/4)} = \sum_{n \text{ even}} \frac{g_n^2}{f_q - f_n}. \tag{4}$$

For typical circuit QED parameters ($f_0 = 6$ GHz, $f_q = 5$ GHz, $g/2\pi = 100$ MHz), we estimate the individual even-mode contributions (Table I).

TABLE I. Even-mode Lamb shift contributions for $f_0 = 6$ GHz, $f_q = 5$ GHz, $g_n \approx g_0\sqrt{f_0/f_n}$.

n	f_n (GHz)	$g_n/2\pi$ (MHz)	$g_n^2/(f_q{-}f_n)$ (kHz)
2	12	71	-714
4	24	50	-132
6	36	41	-54
8	48	35	-29
Total $\Delta(\delta f)$			$\approx -930\,\mathrm{kHz}$

The predicted difference $|\Delta(\delta f)| \approx 930$ kHz ≈ 1 MHz is large compared to the spectroscopic resolution achievable with Ramsey or echo techniques (~ 1 Hz with one hour of averaging [7]). The signal-to-noise ratio exceeds 10^5.

EXPERIMENTAL PROTOCOL

1. **Sample preparation.** Fabricate (or select from existing stock) a $\lambda/2$ and a $\lambda/4$ superconducting resonator with matched fundamental frequencies. Standard aluminum coplanar waveguide or coaxial stub resonators suffice. Couple each to a transmon qubit with $g/2\pi \sim 100$ MHz.
2. **Cooldown.** Mount in a dilution refrigerator at $T \lesssim 20$ mK.
3. **Characterization.** Measure $S_{21}(f)$ to verify mode frequencies:
   - $\lambda/2$: peaks at $f_0, 2f_0, 3f_0, \ldots$
   - $\lambda/4$: peaks at $f_0, 3f_0, 5f_0, \ldots$ (even modes absent)
4. **Qubit spectroscopy.** Measure f_q in each configuration via two-tone spectroscopy or Ramsey interferometry.
5. **Comparison.** Extract $\Delta(\delta f) = f_q^{(\lambda/2)} - f_q^{(\lambda/4)}$ and compare with the theoretical prediction from Eq. (4).

If the same physical qubit can be coupled to both resonators sequentially (e.g., via a tunable coupler or by physically remounting), systematic errors from qubit-to-qubit variation are eliminated.

WHAT THIS MEASUREMENT TESTS

Multi-mode vacuum contribution

The measurement directly quantifies how much of the qubit's Lamb shift comes from higher (even) harmonics. This is relevant for:

- **Frequency targeting**: predicting the dressed qubit frequency in multi-mode environments.
- **Cross-talk**: understanding mode-mediated interactions between qubits in bus architectures.
- **Decoherence**: multi-mode Purcell decay rates depend on the same couplings g_n that produce the Lamb shift.

Completeness of the mode-sum description

Standard circuit QED predicts Eq. (4) as a simple sum of individual even-mode contributions. If the measured $\Delta(\delta f)$ deviates from this prediction, it would indicate a collective or non-perturbative vacuum effect beyond the single-mode-sum picture.

Such a deviation is not expected by standard theory, but *has never been explicitly tested*. This measurement provides the first direct comparison of vacuum fluctuation effects in resonators with structurally different mode spectra.

RELATION TO PRIOR WORK

Lamb shifts and dispersive shifts in circuit QED are well studied [3–5], and multi-mode effects have been analyzed theoretically [8, 9]. Measurements in 3D cavities [6] and multi-mode environments [10] have characterized individual mode couplings.

The novelty of the present proposal is the *controlled comparison* between two resonators differing only in mode parity—one containing all harmonics, the other only odd harmonics—providing a clean test of the vacuum mode sum.

We note that the mode structure of a $\lambda/4$ resonator (odd harmonics only: $1, 3, 5, \ldots$) is equivalent to the mode structure of a cavity with Dirichlet–Neumann (DN) mixed boundary conditions, whose vacuum energy has the opposite sign to the standard Dirichlet–Dirichlet (DD) case—a result first obtained by Boyer [11]. The present measurement can thus be viewed as a circuit QED realization of Boyer's fifty-year-old prediction, probing whether the different mode-parity structure leads to measurably different vacuum physics.

CONCLUSION

We propose comparing the Lamb shift of a transmon qubit in $\lambda/4$ and $\lambda/2$ superconducting resonators. The predicted difference ($\sim 1\,\mathrm{MHz}$) is five orders of magnitude above the measurement floor. The experiment requires no new hardware and can be performed with standard components available in any superconducting circuit laboratory. A positive result validates the multi-mode vacuum contribution to qubit frequency; any deviation from the single-mode-sum prediction would reveal collective vacuum effects not captured by standard perturbation theory.

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